

Los Angeles Crime Analytics

P R E S E N T A T I O N

An End-to-End Power BI Business Intelligence Solution

P R E P A R E D B Y

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Business Problem

The Challenge

- Large volumes of crime data are difficult to analyze in spreadsheets or static reports.
- Identifying patterns, geographic hotspots, and long-term trends is time-consuming.
- Limited visibility slows effective decision-making.



Crime Data



Geographic Hotspots



Interactive Dashboard



Actionable Insights

Project Response

- Develop an interactive Power BI dashboard.
- Transform raw crime records into meaningful insights.

Project Objectives

Transform raw crime data into an interactive Business Intelligence solution for data-driven analysis.

- Analyze reported crime incidents from 2020–2024.
- Identify geographic crime hotspots.
- Compare crime trends over time.
- Build an interactive Power BI dashboard.
- Deliver actionable business insights.



Data Preparation & Transformation

Data Cleaning Process

- Imported raw data.
- Renamed columns.
- Removed duplicates.
- Handled missing values.
- Standardized data types.
- Removed unnecessary columns.

Why Power Query?

- Cleans and transforms raw records before loading them into the Power BI data model.
- Improves data quality, model performance, and reporting accuracy.



Import Raw Data



Clean & Dedupe



Load to Power BI

Dataset Overview

DATA SOURCE

Los Angeles Open Data Portal

DATASET

Crime Data from 2020 to Present

ANALYSIS PERIOD

2020 – 2024

KEY DATA FIELDS

- Case Number
- Report Date
- Crime Category
- Reporting Area
- Victim Information
- Weapon Information
- Status / Clearance
- Geographic Location

PURPOSE

The dataset provides incident-level crime records used to analyze temporal, geographic, demographic, and crime-related patterns across Los Angeles.

CASE_NUMBER	Date Rptd	DATE OCC	TIME OCC	COD_AREA	Rpt Dist No	Part 1-2	Crn Cd	Vict Age	Vict Sex	Vict Descent	Premis Cd	Weapon Used Cd	Status
211507896	4/11/2021	11/7/2020	845	15	1502	2	354	31	M	H	501	-9	IC
201516622	10/21/2020	10/18/2020	1845	15	1521	1	230	32	M	H	102	200	IC
240913563	12/10/2024	10/30/2020	1240	9	933	2	354	30	M	W	501	-9	IC
210704711	12/24/2020	12/24/2020	1310	7	782	1	331	47	F	A	101	-9	IC
201418201	10/3/2020	9/29/2020	1830	14	1454	1	420	63	M	H	103	-9	IC
240412063	12/11/2020	11/11/2020	1210	4	429	2	354	35	M	B	502	-9	IC
240317069	12/16/2024	4/16/2020	1350	3	396	2	354	21	F	B	501	-9	IC
201115217	10/29/2020	7/7/2020	1400	11	1133	2	812	14	F	H	121	500	AO
241708596	4/20/2024	3/2/2020	1200	17	1729	2	354	43	M	W	501	-9	IC
242113813	12/18/2024	9/1/2020	900	21	2196	2	354	57	M	W	501	-9	IC
240605846	2/6/2024	6/26/2020	1	6	657	2	812	13	M	H	501	400	AA
242014110	12/18/2024	11/17/2020	1320	20	2023	2	354	34	F	A	502	-9	IC
202113531	9/6/2020	9/5/2020	1500	21	2149	1	510	0			108	-9	AA
201710725	7/3/2020	7/2/2020	500	17	1762	1	310	0	X	X	203	-9	IC
201406733	2/16/2020	2/13/2020	2300	14	1406	1	330	32	M	W	707	-9	IC
201405970	2/1/2020	2/1/2020	1658	14	1494	1	440	39	M	O	212	-9	AO
201820230	11/8/2020	11/8/2020	730	18	1844	2	626	26	F	B	101	400	IC
201707577	3/20/2020	3/20/2020	1320	17	1752	2	624	21	F	W	210	500	IC

Data Modeling & DAX

Data Modeling:

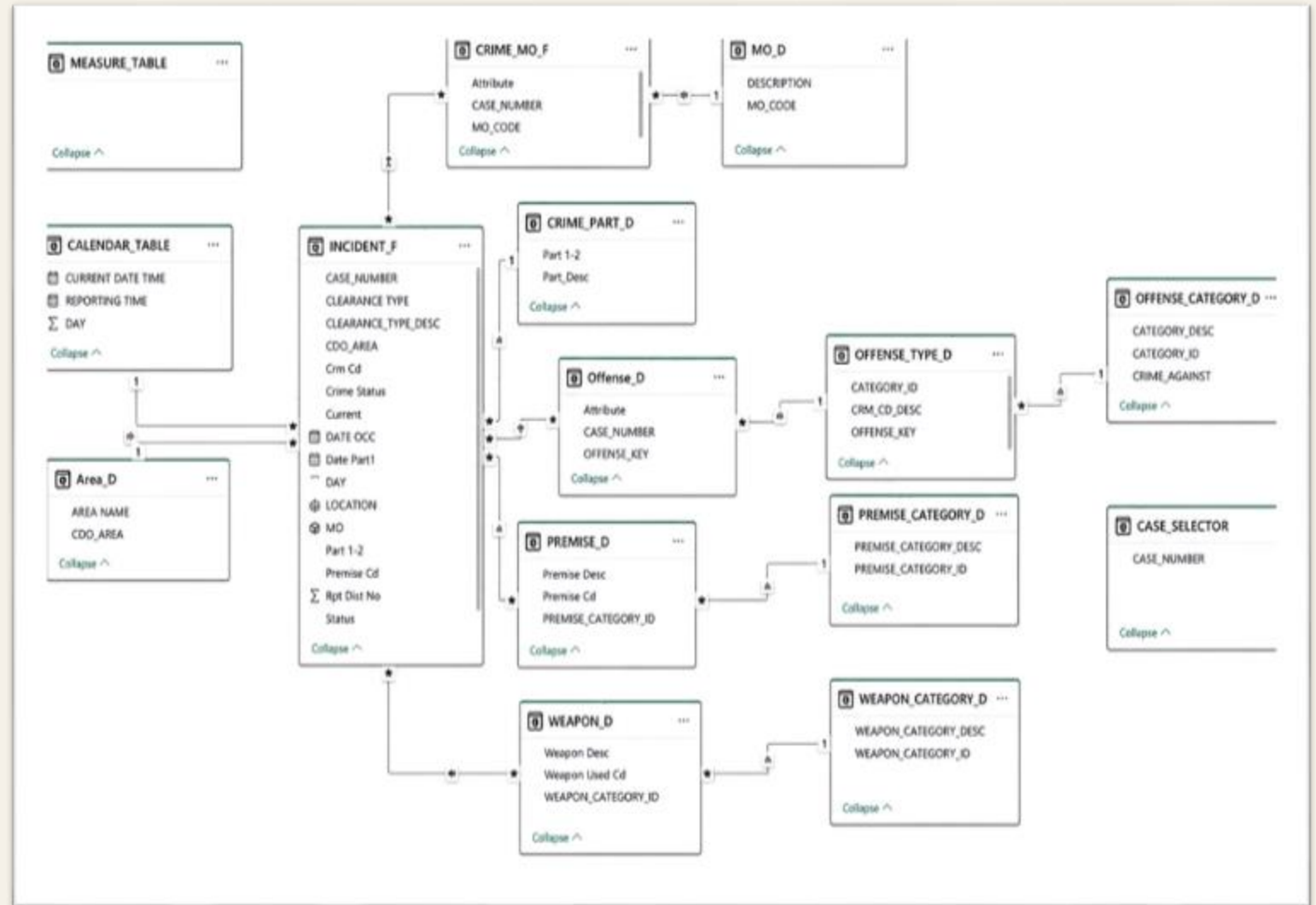
- ✓ Star Schema
- ✓ Fact & Dimension Tables
- ✓ Table Relationships
- ✓ Optimized Data Structure
- ✓ Efficient Cross-Filtering

DAX Development:

- ✓ KPI Measures
- ✓ Calculated Measures
- ✓ Calculated Columns
- ✓ Dynamic Filtering Logic
- ✓ Interactive Analysis

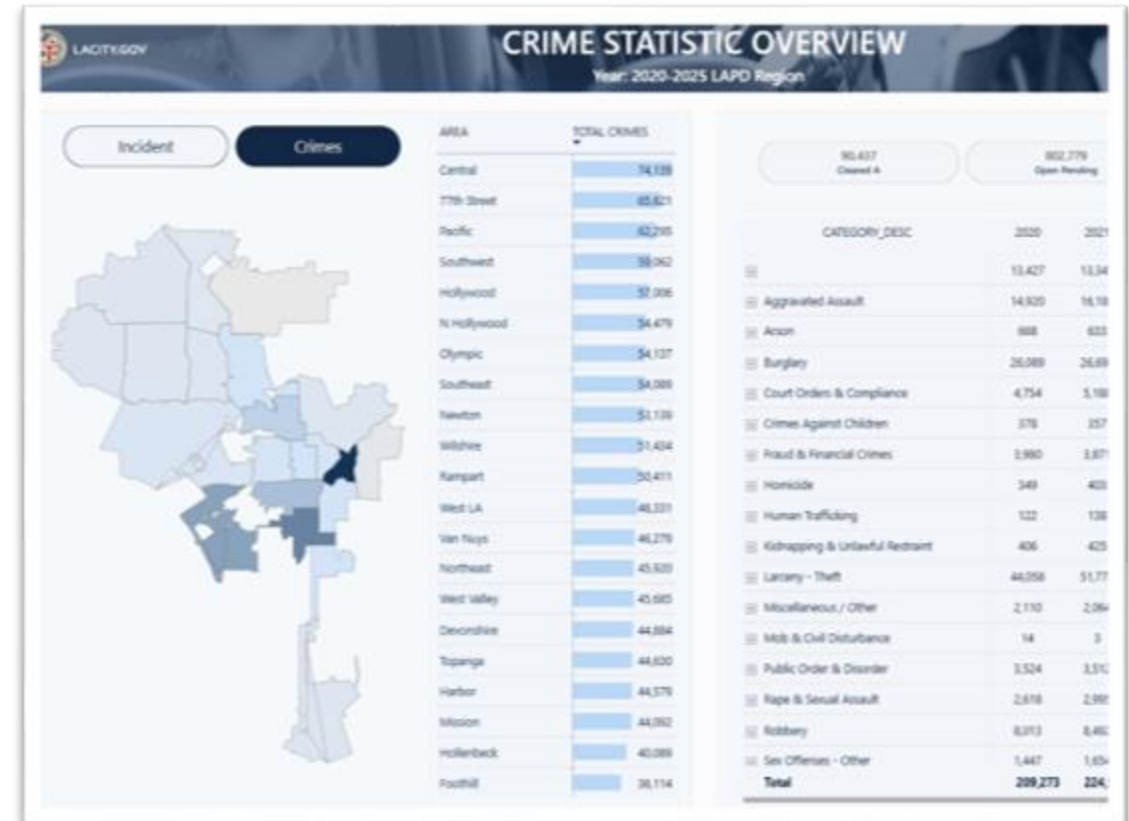
Why Data Modeling?

A structured data model was created to connect fact and dimension tables, improve filtering efficiency, and support DAX calculations used throughout the interactive Power BI dashboard.



Dashboard Features

- **Interactive LAPD Division Map**
Explore crime activity geographically across LAPD reporting divisions.
- **Dynamic KPI Cards**
Display key crime metrics that update based on user selections.
- **Crime Trend Analysis**
Analyze how reported crime changes over time.
- **Yearly Comparisons**
Compare crime patterns and performance across multiple years.
- **Monthly & Weekly Heatmaps**
Identify recurring temporal patterns and periods of higher crime activity.
- **Interactive Filters**
Filter the dashboard dynamically by crime, location, and time.
- **Incident & Crime Analysis Pages**
Explore individual incidents and crime patterns in greater detail.



Key Insights

Geographic Hotspots

- Identify LAPD divisions with the highest reported crime volume.

Crime Trends

- Monitor changes in reported crime from 2020–2024.

Seasonal Patterns

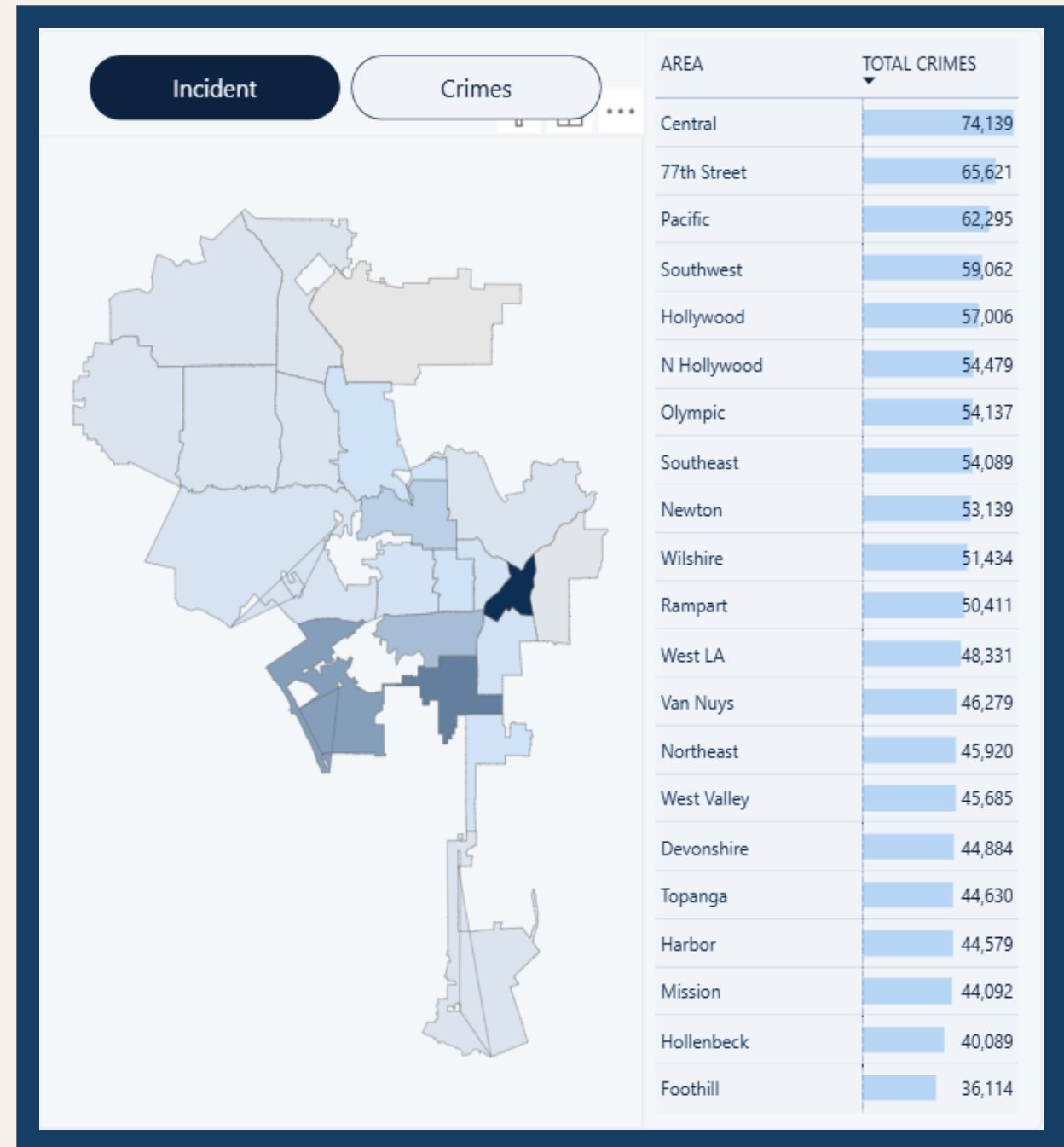
- Detect monthly and weekly patterns in reported incidents.

Crime Category Comparison

- Compare crime categories and their patterns across different years.

Interactive Investigation

- Explore individual incidents using dynamic filters and drill-down analysis.



Skills Demonstrated

Data Preparation

Power Query
Data Cleaning
Data Transformation

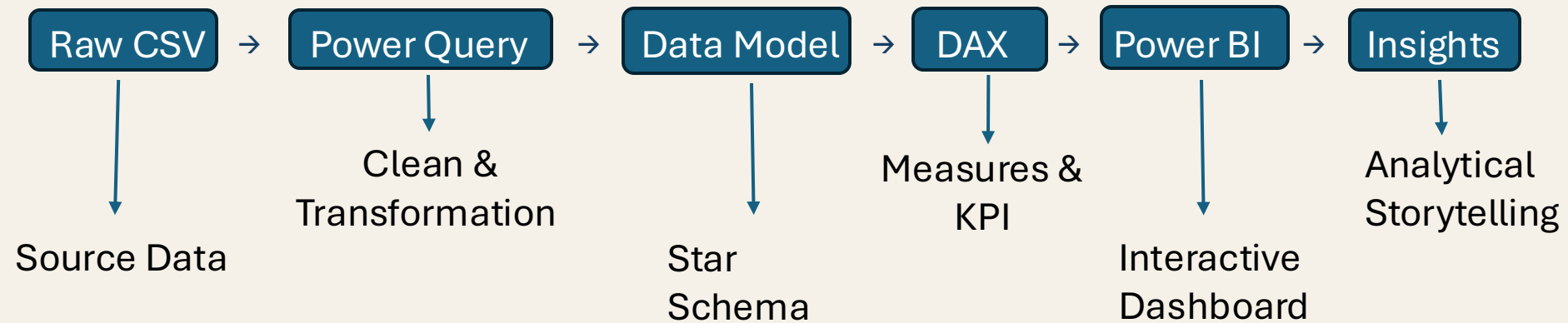
Analytics & Modeling

DAX
Data Modeling
Business Intelligence

Visualization & BI

Power BI
Dashboard Design
Data Visualization

APPLIED THROUGHOUT THE PROJECT



Conclusion

01 BUILT

End-to-End Analytics Solution

Prepared and transformed raw data, developed a structured data model, created DAX measures, and designed an interactive Power BI dashboard.

02 DELIVERED

Clear & Accessible Insights

Enabled users to explore crime trends, geographic patterns, offense categories, and individual incidents through dynamic filtering and visual analysis.

03 DEMONSTRATED

Practical Data Analyst Skills

Applied data cleaning, dimensional modeling, analytical thinking, visualization, and business intelligence techniques within a complete portfolio project.

KEY TAKEAWAY | Effective analytics is not just about building dashboards, it is about transforming complex data into information, people can understand, explore, and use.